Paving Products — Single Drum Vibratory Compactors (cont'd)

Model	Product Ident. No. Prefix	Years Built	Flywheel kW (hp)	Approx. Op. Weight kg (lb)	Drive	Drum Width mm (in)	Centrifugal Force kg (lb)	Maximum Op. Speed km/h (mph)
CS-573C	6LN	95-00	108	13 800	Wheel/	2134	24 091	12.8
			(145)	**(30,360)**	Drum	**(84)**	**(53,000)**	**(8.0)**
CS-573D	CMK	00-02	108	12 180	Wheel/	2130	27 216	12.7
			(145)	**(29,060)**	Drum	**(84)**	**(60,000)**	**(7.8)**
CS64		07-13	116	14 238	Wheel/	2134	28 712	11.4
			(156)	**(31,389)**	Drum	**(84)**	**(63,300)**	**(7.0)**
CP64		07-13	116	14 311	Wheel/	2134	28 712	11.4
			(156)	**(31,550)**	Drum	**(84)**	**(63,300)**	**(7.0)**
CS-583	8YJ	91-95	108	15 040	Wheel/	2134	22 680	12.8
			(145)	**(33,090)**	Drum	**(84)**	**(50,000)**	**(8.0)**
CS-583C	7MN	95-00	108	15 230	Wheel/	2134	24 091	12.8
			(145)	**(33,500)**	Drum	**(84)**	**(53,000)**	**(8.0)**
CS-583D	2CZ	98-02	108	14 850	Wheel/	2130	31 751	12.7
			(145)	**(32,740)**	Drum	**(84)**	**(70,000)**	**(7.8)**
CS74		07-13	116	15 455	Wheel/	2134	33 840	11.4
			(156)	**(34,072)**	Drum	**(84)**	**(74,600)**	**(7.0)**
CP74		07-13	116	15 333	Wheel/	2134	33 840	11.4
			(156)	**(33,804)**	Drum	**(84)**	**(74,600)**	**(7.0)**
CS-643	7FD	85-87	100	14 900	Wheel/	2200	16 800	15.5
			(134)	**(32,855)**	Drum	**(86)**	**(37,044)**	**(9.6)**
CP-643	7GD	85-87	100	16 300	Wheel/	2200	12 600	15.5
			(134)	**(35,942)**	Drum	**(86)**	**(27,783)**	**(9.6)**
CS76		07-13	130	16 758	Wheel/	2134	33 840	11.4
			(174)	**(36,945)**	Drum	**(84)**	**(74,600)**	**(7.0)**
CP76		07-13	130	16 896	Wheel/	2134	33 840	11.4
			(174)	**(37,249)**	Drum	**(84)**	**(74,600)**	**(7.0)**
CS-653	7HD	85-91	100	17 100	Wheel/	2200	22 230	15.5
			(134)	**(37,690)**	Drum	**(86)**	**(48,995)**	**(9.6)**
CS76 XT		07-13	130	18 611	Wheel/	2134	33 840	11.4
			(174)	**(41,030)**	Drum	**(84)**	**(74,600)**	**(7.0)**
CP-653	7JD	85-91	100	18 500	Wheel/	2200	22 230	15.5
			(134)	**(40,774)**	Drum	**(86)**	**(48,995)**	**(9.6)**
TSF-54	7KD	85-88	26	2131	Towed	1370	6810	Towed
			(35)	**(4700)**		**(54)**	**(15,000)**	
TSM-54	7LD	86-88	26	2160	Towed	1370	6810	Towed
			(35)	**(4760)**		**(54)**	**(15,000)**	



PAVING PRODUCTS — DOUBLE DRUM, COMBI AND PNEUMATIC TIRE COMPACTORS

Model	Product Ident. No. Prefix	Years Built	Flywheel kW (hp)	Approx. Op. Weight kg (lb)	Drive	Drum Width mm (in)	Centrifugal Force kg (lb)	Maximum Op. Speed km/h (mph)
CB14	DTT	07-12	16.1 **(21.6)**	1620 **(3571)**	Hydraulic	800 **(31)**	1051 **(2318)**	8.5 **(5.0)**
CB14 XW	DTT	07-12	16.1 **(21.6)**	1840 **(4057)**	Hydraulic	1000 **(39)**	1164 **(2565)**	8.5 **(5.0)**
CB14 Full Flush	DTT	07-12	16.1 **(21.6)**	1600 **(3527)**	Hydraulic	900 **(35)**	1051 **(2318)**	8.5 **(5.0)**
CB-214	6FD	85-88	24 **(33)**	2300 **(5070)**	Drum (2)	1000 **(39.4)**	2041 **(4500)**	10.6 **(6.6)**
CB-214B	6LF	88-93	24 **(33)**	2300 **(5072)**	Drum (2)	990 **(39)**	2018 **(4450)**	10.4 **(6.5)**
CB-214C	6LF	93-00	25 **(33)**	2320 **(5115)**	Drum (2)	1000 **(39.4)**	2592 **(5715)**	10.5 **(6.5)**
CB-214D	1TZ	99-03	23.5 **(31.5)**	2430 **(5355)**	Hydraulic	1000 **(39.4)**	2590 **(5710)**	10.0 **(6.2)**
CB-214E		-08	24.4 **(32.7)**	2450 **(5400)**	Hydraulic	1000 **(39)**	2751 **(6075)**	10.0 **(6.0)**
CB-224	6GD	85-88	24 **(33)**	2450 **(5400)**	Drum (2)	1200 **(47.2)**	2450 **(5400)**	10.6 **(6.6)**
CB-224B	6LF	88-93	24 **(33)**	2450 **(5402)**	Drum (2)	1199 **(47.2)**	2449 **(5400)**	10.4 **(6.5)**
CB-224C	6LF	93-00	25 **(33)**	2420 **(5335)**	Drum (2)	1200 **(47.2)**	2920 **(6570)**	10.5 **(6.5)**
CB-224D	8RZ	99-03	23.5 **(31.5)**	2610 **(5750)**	Hydraulic	1200 **(47.2)**	3039 **(6700)**	10.0 **(6.2)**
CB22		08-13	24.6 **(33)**	2500 **(5512)**	Hydraulic	1000 **(39)**	2817 **(6210)**	10.5 **(6.5)**
CB24		08-13	24.6 **(33)**	2700 **(5952)**	Hydraulic	1200 **(47)**	3195 **(7043)**	10.5 **(6.5)**
CB32		08-13	24.6 **(33)**	3230 **(7121)**	Hydraulic	1300 **(51)**	3195 **(7043)**	10.5 **(6.5)**
CC24		08-13	24.6 **(33)**	2400 **(5291)**	Hydraulic	1200 **(47)**	3195 **(7043)**	10.5 **(6.5)**
CB-224E		-08	24.4 **(32.7)**	2630 **(5800)**	Hydraulic	1200 **(47)**	3164 **(6975)**	10.0 **(6.0)**
CB-225D	9FZ	99-03	23.5 **(31.5)**	2390 **(5270)**	Hydraulic	1200 **(47.2)**	3039 **(6700)**	10.0 **(6.2)**
CB-225E		-08	24.4 **(32.7)**	2300 **(5070)**	Hydraulic	1200 **(47)**	3164 **(6975)**	10.0 **(6.0)**
CB-314	6HD	85-89	41 **(55)**	3357 **(7400)**	Drum	1120 **(44)**	2770 **(6100)**	8.0 **(5.0)**
CB-334D	3JZ 4CZ DCZ	98-03 98-02 02-04	32 **(43)**	3850 **(8490)**	Hydraulic	1300 **(51)**	3263 **(7250)**	11.0 **(7.0)**
CB-334E		-08	34.1 **(46)**	3940 **(8688)**	Hydraulic	1300 **(51)**	3378 **(7448)**	12.5 **(8.0)**

Paving Products — Double Drum, Combi and Pneumatic Tire Compactors (cont'd)

Model	Product Ident. No. Prefix	Years Built	Flywheel kW (hp)	Approx. Op. Weight kg (lb)	Drive	Drum Width mm (in)	Centrifugal Force kg (lb)	Maximum Op. Speed km/h (mph)
CB-335D	5PZ	98-03	32	3620	Hydraulic	1300	3263	11.0
	BBW	00-03	**(43)**	**(7980)**		**(51)**	**(7250)**	**(7.0)**
	D4E	02-03						
CB34		07-13	34.1	3940	Hydraulic	1300	3378	12.5
			(46)	**(8688)**		**(51)**	**(7448)**	**(8.0)**
CC34		07-13	34.1	3670	Hydraulic	1300	3378	12.5
			(46)	**(8091)**		**(51)**	**(7448)**	**(8.0)**
CB-335E		-08	34.1	3670	Hydraulic	1300	3378	12.5
			(46)	**(8092)**		**(51)**	**(7448)**	**(8.0)**
CB-414	6KD	85-89	52	5780	Drum	1397	6350	13.7
			(70)	**(12,750)**		**(55)**	**(14,000)**	**(8.5)**
CB-424	6LD	85-89	54	6220	Drum (2)	1397	4485	11.0
			(73.5)	**(13,710)**		**(55)**	**(9885)**	**(6.8)**
CB-434	3TF	89-94	60	6610	Drum (2)	1422	7620	11.6
			(80)	**(14,540)**		**(56)**	**(16,800)**	**(7.2)**
CB-434B	6AL	94-95	60	6577	Drums	1422	7620	0-11.6
			(80)	**(14,500)**		**(56)**	**(16,800)**	**(0-7.2)**
CB-434C	4DN	95-03	52	6485	Hydraulic	1422	7620	11.6
			(70)	**(14,300)**		**(56)**	**(16,800)**	**(7.2)**
CB-434D		04-11	62	7500	Hydraulic	1500	7960	11.6
			(83)	**(16,535)**		**(59)**	**(17,550)**	**(7.0)**
CB-434D XW		04-11	62	7700	Hydraulic	1700	7960	11.6
			(83)	**(16,975)**		**(67)**	**(17,550)**	**(7.0)**
CB-514	6YD	85-88	68	9730	Drum (2)	1730	9073	11.0
			(91)	**(21,450)**		**(68)**	**(20,000)**	**(7.0)**
CB-521	6RD	85-87	61	8800	Wheel	1700	5300	15.0
			(82)	**(19,404)**		**(67)**	**(11,687)**	**(9.3)**
CB-522	6SD	85-87	45	10 100	Drum (2)	1700	10 350	8.0
			(62)	**(22,271)**		**(67)**	**(22,822)**	**(5.0)**
CB-523	6TD	85-87	61	8800	Wheel/	1700	5300	13.0
			(82)	**(19,404)**	Drum	**(67)**	**(11,687)**	**(8.0)**
CB-524	6WD	85-87	61	9500	Drum (2)	1700	10 350	11.0
			(82)	**(20,948)**		**(67)**	**(22,822)**	**(6.8)**
CB-534	6EG2YF	87-93	93	9117	Drum (2)	1700	11 800	11.2
			(125)	**(20,100)**		**(67)**	**(26,019)**	**(7.0)**
CB-534B	4JL	93-95	80	9117	Drums	1676	12 043	0-11.3
			(107)	**(20,100)**		**(66)**	**(26,550)**	**(0-7.0)**
CB-534C	5HN	95-02	75	9195	Drums	1700	11 975	11.3
			(100)	**(20,270)**		**(67)**	**(26,400)**	**(7.0)**
CB-534D			97	10 380	Drums	1700	11 434	13.0
			(130)	**(22,836)**		**(67)**	**(25,208)**	**(8.0)**
CB54		09-12	102	10 804	Hydraulic	1700	11 300	13.0
			(137)	**(23,818)**		**(67)**	**(24,929)**	**(8.0)**
CB-534D XW			97	11 300	Drums	2000	11 434	13.0
			(130)	**(24,860)**		**(79)**	**(25,208)**	**(8.0)**
CB54 XW	N/A		102	11 898	Hydraulic	2000	11 308	13.0
			(137)	**(26,230)**		**(6'7")**	**(24,929)**	**(8.0)**

Paving Products — Double Drum, Combi and Pneumatic Tire Compactors (cont'd)

Model	Product Ident. No. Prefix	Years Built	Flywheel kW (hp)	Approx. Op. Weight kg (lb)	Drive	Drum Width mm (in)	Centrifugal Force kg (lb)	Maximum Op. Speed km/h (mph)
CB-544	8FM	94-01	60 **(80)**	10 700 **(23,593)**	Drums	1700 **(67)**	8850 **(19,510)**	8.9 **(5.5)**
CB-545	2FS	96-01	60 **(80)**	9410 **(20,750)**	Wheel/ Drum	1700 **(67)**	8850 **(19,510)**	8.9 **(5.5)**
CB-564D			97 **(130)**	12 600 **(27,783)**	Drums	2130 **(84)**	10 085 **(22,234)**	13.0 **(8.0)**
CB-614	7CD	85-93	115 **(155)**	11 340 **(25,000)**	Drum (2)	1980 **(78)**	9525 **(21,000)**	11.2 **(7.0)**
CB-634	5CL	94-95	108 **(145)**		Drums	2134 **(84)**	12 043 **(26,550)**	0-9.2 **(0-5.7)**
CB-634C	3BR	95-01	108 **(145)**	11 680 **(25,750)**	Drums	2134 **(84)**	12 043 **(26,550)**	12.2 **(7.6)**
CB64	N/A		102 **(137)**	12 980 **(28,616)**	Hydraulic	1680 **(5'6")**	14 107 **(31,100)**	13.0 **(8.0)**
CD54		09-12	74.5 **(99.5)**	9500 **(20,944)**	Hydraulic	1700 **(67)**	8573 **(18,900)**	9.0 **(5.5)**
PF-200		85-92	49 **(66)**	7000 **(15,430)**	Wheel Pneumatic	1700 **(67)**	N/A	24.0 **(14.9)**
PS-110	7MD	85-96	57 **(77)**	12 500 **(27,550)**	Wheel Pneumatic	2134 **(84)**	N/A	38.6 **(24.0)**
PS-130	7ND	85-96	57 **(77)**	12 500 **(27,550)**	Wheel Pneumatic	1700 **(67)**	N/A	38.6 **(24.0)**
PS-150	7PD	85-96	57 **(77)**	15 050 **(37,300)**	Wheel Pneumatic	1700 **(67)**	N/A	38.6 **(24.0)**
PS-150B	3XR	95-04	52 **(70)**	12 940 **(28,535)**	Hydraulic	1743 **(69)**	N/A	25.6 **(15.9)**
PS150C	N/A		75 **(100)**	12 940 **(28,535)**	Hydraulic	1740 **(5'8")**	NA	25.6 **(15.9)**
PS-180	7PD	85-96	57 **(77)**	16 950 **(37,000)**	Wheel Pneumatic	1727 **(68)**	N/A	38.6 **(24.0)**
PS-200B	5JR	95-04	78 **(105)**	13 010 **(28,685)**	Hydraulic	1743 **(69)**	N/A	19.3 **(12.0)**
PS-300	7TD	85-95	77	21 000	Wheel	1900	N/A	26.5
PF-300	7WD	85-95	**(102)**	**(46,200)**	Pneumatic	**(75)**		**(16.4)**
PS-300B			74	14 000	Wheel	1920	N/A	19.0
PF-300B			**(105)**	**(30,860)**	Pneumatic	**(77)**		**(11.8)**
PF-300C			75	21 000	Wheel	1900		13.0
PS-300C			**(100)**	**(46,200)**	Pneumatic	**(75)**		**(8.0)**
PS-360B			78 **(105)**	8500 **(18,740)**	Wheel Pneumatic	2275 **(90)**	N/A	18.0 **(11.0)**



SURFACE DRILLS — TRACK DRILLS

Model	Hole Diameter	Hole Depth	Air Compressor	Rock Drill	Engine	Years Built
MD5050	• Up to 102 mm **(4.0 in)**	• Up to 31.2 m **(102.5 ft)**	• 7.08 m³/min @ 10.2 bar **(250 ft³/min @ 150 psi)**	• HPR4519 19 kW **(25 hp)**	• 4HK* 150 kW **(203 hp)** at 2100 RPM	Up to 2016
MD5050T	• Up to 102 mm **(4.0 in)**	• Up to 31.2 m **(102.5 ft)**	• 7.08 m³/min @ 10.2 bar **(250 ft³/min @ 150 psi)**	• HPR4519 19 kW **(25 hp)**	• 4HK* 150 kW **(203 hp)** at 2100 RPM	Up to 2016
MD5075	• Up to 127 mm **(5 in)**	• Up to 31.2 m **(102.5 ft)**	• 9.78 m³/min @ 10.2 bar **(350 ft³/min @ 150 psi)**	• HPR4519 19 kW **(25 hp)**	• Cat C9 ACERT* 224 kW **(300 hp)** at 1800 RPM	Up to Aug. 2014
MD5090	• Up to 127 mm **(5 in)**	• Up to 22.1 m **(72.5 ft)**	• 8.5 m³/min @ 10.2 bar **(300 ft³/min @ 150 psi)**	• HPR4519 19 kW **(25 hp)**	• Cat C9 ACERT* 224 kW **(300 hp)** at 1800 RPM	Up to 2016
MD5125	• Up to 152 mm **(6 in)**	• Up to 30 m **(98.5 ft)**	• 11.3 m³/min @ 7 bar **(400 ft³/min @ 100 psi)**	• HPR5128 28 kW **(37 hp)**	• Tier 3 Cat C11 242 kW **(325 hp)** at 1800 RPM	Up to Aug. 2014

*This product emits equivalent to Tier 3, Stage IIIA, and Japan 2006 (Tier 3).



SURFACE DRILLS — ROTARY DRILLS

Model	Hole Diameter	Hole Depth	Air Compressor	Engine	Years Built
MD6240	• Up to 244 mm **(9.625 in)**	• Depth single-pass (12 m mast) — Up to 12.80 m **(42 ft)** • Depth multi-pass — Up to 55.47 m **(182 ft)** • Depth single-pass (15 m mast) — Up to 15.85 m **(52 ft)** • Depth multi-pass — Up to 31.08 m **(102 ft)**	• 48 m^3/min **(1700 ft^3/min)** 6.9 bar **(100 psi)** • 32.6 m^3/min **(1150 ft^3/min)** 10.3/24.1 bar **(150/350 psi)** • 38.3 m^3/min **(1350 ft^3/min)** 10.3/24.1 bar **(150/350 psi)** • 43 m^3/min **(1500 ft^3/min)** 34.5 bar **(500 psi)**	• C27 ACERT* • 597 kW **(800 hp)** at 2100 RPM • 652 kW **(875 hp)** at 2100 RPM	Up to 2016
MD6420	• Up to 311 mm **(12.25 in)**	• Depth single pass (10 m mast) — Up to 10.29 m **(33 ft 9 in)** • Depth multi-pass — Up to 63.4 m **(208 ft)** • Depth single pass (13 m mast) — Up to 13.4 m **(44 ft)** • Depth multi-pass — Up to 74.4 m **(244 ft)** • Depth single pass (16 m mast) — Up to 16.46 m **(54 ft)** • Depth multi-pass — Up to 65.2 m **(214 ft)**	• 67.9 m^3/min **(2400 ft^3/min)** 6.9 bar **(100 psi)** • 42.5 m^3/min **(1500 ft^3/min)** 10.3/24.1/34.4 bar **(150/350/500 psi)** • 67.9 m^3/min **(2400 ft^3/min)** 6.9 bar **(100 psi)**	• C27 ACERT* • 597 kW **(800 hp)** at 2100 RPM • C27 ACERT* • 652 kW **(875 hp)** at 2100 RPM • Cummins QST30* 634 and 783 kW **(850 and 1050 hp)** 783 kW **(1050 hp)** at 2100 RPM	Up to 2016
MD6420B	• Up to 311 mm **(12.25 in)**	• Depth single pass (10 m mast) — Up to 10.29 m **(33 ft 9 in)** • Depth multi-pass — Up to 63.40 m **(208 ft)** • Depth single pass (13 m mast) — Up to 13.4 m **(44 ft)** • Depth multi-pass — Up to 74.4 m **(244 ft)** • Depth single pass (16 m mast) — Up to 16.46 m **(54 ft)** • Depth multi-pass — Up to 65.2 m **(214 ft)**	• 56.6 m^3/min **(2000 ft^3/min)** 6.9 bar **(100 psi)** • 67.9 m^3/min **(2400 ft^3/min)** 6.9 bar **(100 psi)** • 42.5 m^3/min **(1500 ft^3/min)** 34.4 bar **(500 psi)**	• 627 kW **(850 hp)** at 1800 RPM • QST30C* (non-cert with cense) 786 kW **(1050 hp)** at 2100 RPM • QST30* 627 kW **(850 hp)** at 1800 RPM • AST30C (non-cert with cense) 783 kW **(1050 hp)** at 2100 RPM	Up to 2016
MD6540	• Up to 381 mm **(15 in)**	• Depth single-pass (16 m mast) — Up to 16.15 m **(53 ft)** • Depth multi-pass — Up to 84.73 m **(278 ft)** • Depth single pass (20 m mast) • Depth multi-pass — Up to 65.7 m **(215.6 ft)**	• 102 m^3/min **(3600 ft^3/min)** 6.9 bar **(100 psi)**	• MTU Detroit Diesel Series 2000 16V* • 899 kW **(1205 hp)** at 1800 RPM	Up to 2016
MD6750	• Up to 444 mm **(17.5 in)**	• 19.81 m **(65 ft)**	• 107.6 m^3/min **(3800 ft^3/min)**	• AC Induction Motor • Dual Shaft • 597 kW **(800 hp)**	Up to 2014

*This product emits equivalent to Tier 2.



UNDERGROUND MINING — HARD ROCK LOAD-HAUL-DUMPS (LHD'S)

LHD Model	Product Ident. No. Prefix (USA)	Years Built	Flywheel Power kW (hp)	Approx. Shipping Weight kg (lb)	Max. Capacity kg (lb)	Length m (ft)	Height m (ft)	Bucket Width mm (ft)	Breakout Force kg (lb)	Maximum Speeds km/h (mph)	
										Forward	Reverse
R1300	6QW	96-03	123 **(165)**	20 150 **(44,430)**	6500 **(14,330)**	8.66 **(28'5")**	2.00 **(6'7")**	2000 **(6'7")**	12 020 **(26,500)**	26.1 **(16.2)**	23.8 **(14.6)**
R1300G	LJB	03-13	123 **(165)**	20 950 **(46,187)**	6800 **(14,991)**	8.71 **(28'7")**	2.12 **(6'11")**	2200 **(7'3")**	12 020 **(26,500)**	24.0 **(14.9)**	23.0 **(14.3)**
R1300GII	RSL	06-08	136 **(182)**	20 875 **(46,021)**	6800 **(14,991)**	8.71 **(28'7")**	2.12 **(6'11")**	2200 **(7'3")**	15 350 **(33,841)**	26.1 **(16.2)**	25.9 **(16.1)**
R1500	BAY	86-00	178 **(239)**	25 100 **(55,360)**	9000 **(19,850)**	9.19 **(30'2")**	2.30 **(7'7")**	2480 **(8'2")**	18 460 **(40,700)**	30.4 **(18.9)**	33.0 **(20.5)**
R1600	9EW	97-03	201 **(270)**	29 800 **(65,710)**	10 200 **(22,490)**	9.71 **(31'10")**	2.40 **(7'10")**	2600 **(8'6")**	19 280 **(42,510)**	30.6 **(19.0)**	34.0 **(21.1)**
R1600	9XP	01-03	201 **(270)**	29 800 **(65,710)**	10 200 **(22,490)**	9.71 **(31'10")**	2.40 **(7'10")**	2600 **(8'6")**	19 280 **(42,510)**	30.6 **(19.0)**	34.0 **(21.1)**
R1600G	9PP	03-08	201 **(270)**	29 800 **(65,700)**	10 200 **(22,490)**	9.71 **(31'10")**	2.40 **(7'10")**	2600 **(8'6")**	19 000 **(41,888)**	22.9 **(14.2)**	23.8 **(14.8)**
R1600G	9YZ	07-13	201 **(270)**	29 800 **(65,700)**	10 200 **(22,490)**	9.71 **(31'10")**	2.40 **(7'10")**	2600 **(8'6")**	19 000 **(41,888)**	22.9 **(14.2)**	23.8 **(14.8)**
R1700 Series II	4LZ	94-00	231 **(310)**	34 500 **(76,100)**	12 000 **(26,460)**	10.42 **(34'2")**	2.53 **(8'4")**	2820 **(9'3")**	23 430 **(51,660)**	29.3 **(18.2)**	33.3 **(20.7)**
R1700G	8XZ	99-06	262/241 **(351/323)**	38 500 **(84,878)**	14 000* **(30,865)***	10.59 **(34'9")**	2.56 **(8'5")**	2894 **(9'6")**	20 100 **(44,313)**	24.1 **(15.0)**	25.3 **(15.7)**
R1700G	SBR/XBR	06-19	242/263 **(324/353)**	38 500 **(84,878)**	14 000* **(30,865)***	10.74 **(35'3")**	2.56 **(8'5")**	2772 **(9'1")**	20 885 **(46,051)**	24.1 **(15.0)**	25.3 **(15.7)**
R2800	BBR	88-98	231 **(310)**	42 660 **(94,070)**	16 200 **(35,720)**	10.70 **(35'1")**	2.68 **(8'10")**	3000 **(9'10")**	26 540 **(68,530)**	29.3 **(18.2)**	33.3 **(20.7)**
R2900	5TW	95-03	269 **(361)**	48 850 **(107,710)**	17 200 **(37,920)**	10.97 **(36'0")**	2.89 **(9'6")**	3100 **(10'2")**	28 600 **(63,060)**	26.7 **(16.6)**	32.5 **(20.2)**
R2900G	GLK	02-05	306 **(410)**	50 100 **(110,451)**	17 200 **(37,920)**	11.3 **(37'1")**	2.89 **(9'6")**	3176 **(10'5")**	27 346 **(60,298)**	25.3 **(15.7)**	26.4 **(16.4)**
R2900G XTRA	GLK	02-05	306 **(410)**	53 100 **(117,065)**	20 000* **(44,092)***	11.3 **(37'1")**	2.89 **(9'6")**	3400 **(11'2")**	27 346 **(60,298)**	25.3 **(15.7)**	26.4 **(16.4)**
R2900G XTRA	JLK	05-13	333 **(447)**	55 575 **(122,522)**	20 000* **(44,092)***	11.3 **(37'1")**	2.99 **(9'10")**	3500 **(11'6")**	27 346 **(60,298)**	25.3 **(15.7)**	26.4 **(16.4)**

*Capacity is for tramming only. Truck loading capacity is lower.



UNDERGROUND MINING — HARD ROCK TRUCKS

Articulated Trucks

Truck Model	Product Ident. No. Prefix (USA)	Years Built	Flywheel Power kW (hp)	Approx. Shipping Weight kg (lb)	Max. Capacity kg (lb)	Length m (ft)	Height m (ft)	Body Width mm (ft)	Maximum Speeds km/h (mph)	
									Forward	Reverse
AD30	CXR	02-05	304 **(408)**	28 870 **(63,647)**	30 000 **(66,139)**	10.2 **(33'4")**	2.60 **(8'6")**	2690* **(8'10")***	40.8 **(25.4)**	7.8 **(4.8)**
AD40	1YZ	94-02	380 **(510)**	38 100 **(84,000)**	40 000 **(88,200)**	10.7 **(35'0")**	2.70 **(8'10")**	3000 **(9'10")**	48.1 **(29.9)**	10.6 **(6.6)**
AD40 Series II	N/A	N/A	380 **(510)**	41 800 **(92,170)**	40 000 **(88,200)**	11.3 **(37'0")**	2.89 **(9'6")**	3200 **(10'5")**	41.7 **(25.9)**	9.6 **(6.0)**
AD45	BKZ	01-05	439 **(589)**	39 359 **(86,772)**	45 000 **(99,208)**	11.2 **(36'9")**	2.82 **(9'5")**	3000* **(9'10")***	52.0 **(32.3)**	10.7 **(6.6)**
AD55	ANW	01-03	485 **(650)**	47 000 **(103,617)**	55 000 **(121,254)**	11.6 **(37'11")**	3.20 **(10'6")**	3346* **(11'0")***	42.3 **(26.3)**	9.0 **(5.6)**
AD55B	JNW	07-13	600 **(805)**	50 000 **(110,231)**	55 000 **(121,254)**	12.0 **(39'0")**	3.4 **(11'2")**	3346* **(11'0")***	41.5 **(25.8)**	8.7 **(5.4)**

Rigid Frame Trucks

69D Dump	9XS	00-04	380 **(510)**	30 100 **(66,371)**	38 000 **(83,790)**	8.1 **(26'8")**	3.4 **(11'4")**	3665 **(12'0")**	76.6 **(47.6)**	13.5 **(8.4)**
69D Ejector	9XS	00-04	380 **(510)**	34 700 **(78,514)**	36 200 **(79,821)**	7.8 **(25'8")**	3.4 **(11'4")**	3665 **(12'0")**	75.2 **(46.7)**	13.5 **(8.4)**

*Overall width less body.



UNDERGROUND MINING — ROOM & PILLAR — CONTINUOUS MINERS

Model	Prefix	Exit Year	Bucyrus Model	Total Power kW (hp)	Cutter Head Power kW (hp)	Loading Capacity tonnes/min (Tons/min)	Recommended Mining Range mm (in)	Machine Weight tonnes (lb)
CM235	GEG	2014	25C	701 **(940)**	2 × 205 **(2 × 275)**	14-29 **(15-32)**	914-3251 **(36-128)**	61.2 **(135,000)**
CM330	GEY	2014	30M2	698 **(930)**	2 × 104 **(2 × 280)**	24 **(26.5)**	up to 3170 **(up to 125)**	61.7 **(136,000)**
CM845	GE6	2015	30MB	674 **(904)**	2 × 173 **(2 × 232)**	31.5 **(35)**	2800-3900 **(110-153)**	92.0 **(202,800)**



UNDERGROUND MINING — ROOM & PILLAR — LHDs

Model	Prefix	Exit Year	Bucyrus Model	Bucket Payload kg (lb)	Fork Payload kg (lb)	Engine	Engine Power kW (hp)	Machine Weight kg (lb)
CL106		2014	N/A					
CL110	LH4	2014	N/A	8000 **(17,636)**	10 000 **(22,046)**	Cat C7 ACERT	171.5 **(230)**	21 500 **(47,400)**
CL115	LH6	2014	N/A					
CL215B		2014	MH15					



UNDERGROUND MINING — ROOM & PILLAR — SCOOPS

Model	Prefix	Exit Year	Bucyrus Model	Lift Capacity tonnes (Tons)	Engine	Engine Power kW (hp)	Tram Speed km/h (mph)	Weight with Battery kg (lb)
SU482 DN (diesel non permissible)	LAW	2014	482D	6.8 **(7.5)**	Duetz BF4L 2011	58 **(78)**	16.9 **(10.5)**	
SU482	LAP	2014	488XL	6.4 **(7.0)**	Battery		8.0 **(5.0)**	
SU488 DM (dual motor)	LA4	2014	488-6 DM	14.5 **(16.0)**	Battery		8.0 **(5.0)**	26 853 **(59,200)**
SU488 D XP (explosion proof)		2014	488D XP	14.5 **(16.0)**	Duetz	112 **(150)**	17.9 **(11.1)**	
SU488 LHD	LA6	2014	488D LHD XP	10.9 **(12.0)**	Duetz 1013 FC	112 **(150)**	17.9 **(11.1)**	
SU488 LHD N	LA7	2014	488D LHD	10.9 **(12.0)**	Duetz 1013 FC	112 **(150)**	17.9 **(11.1)**	
SU488 DM AC (alternating current)	LA4	2014	488-6 DM	14.5 **(16.0)**	Battery		8.0 **(5.0)**	26 853 **(59,200)**
SU636		2014						



UNDERGROUND MINING — ROOM & PILLAR — HAULAGE (FACE HAULERS, CONTINUOUS HAULAGE)

Model	Prefix	Exit Year	Bucyrus Model	Payload Capacity tonnes (Tons)	Machine Weight — empty kg (lb)	Minimum Operating Height mm (in)	Engine Type	Engine Power kW (hp)
FH110	FHM	2014	810C Un-a-trac	9.0 **(10.0)**	31 524 — with battery **(69,500)**	1066 **(42)**		
FH125 D	FHX	2014	FBR 15 Ram Car	20.0 **(22.0)**	25 000 — full tanks **(55,115)**	1900 **(74.8)**	Cat 3126	172 **(230)**
FH336 (Continuous Haulage)	FHZ	2014	36 CHH	27.0 **(30.0)**	27 216 **(60,000)**	965 **(38)**		



UNDERGROUND MINING — ROOM & PILLAR — FEEDER BREAKERS

Model	Prefix	Exit Year	Bucyrus Model	Capacity Throughput tonnes/hr (Tons/hr)	Minimum Operating Height mm (in)	Operating Weight kg (lb)	Total Power kW (hp)
FB75	FBP	2014	7MFBH-48A LOW	717 **(790)**	1270 **(50)**	31 751 **(70,000)**	151 **(200)**
			GH-MFBHM-48CDL	1089 **(1200)**	1219 **(48)**	27 215 **(60,000)**	151 **(200)**
FB75 P (permissible)	FBP	2014					
FB85	FBE	2014	7FB-56AL	860 **(950)**	836 **(34)**	31 750 **(70,000)**	150 **(200)**
FB85 P (permissible)	FBE	2014	7FB-56AL	860 **(950)**	837 **(34)**	31 750 **(70,000)**	151 **(200)**
FB110 P (permissible)	FBL	2014	7FB-48A	1146 **(1263)**	1219 **(48)**	31 750 **(70,000)**	151 **(200)**



UNDERGROUND MINING — ROOM & PILLAR — ROOF BOLTERS

Model	Prefix	Exit Year	Bucyrus Model	Drilling Torque N·m (lbf-ft)	Minimum Tram Height mm (ft)	Maximum ATRS Reach mm (ft)
RB120	RM9	2014	LRB-15AR	380 **(280)**	813 **(2'8")**	1930 **(6'4")**
RB230	RP9	2014	RB2-88A	420 **(310)**	1181 **(3'10.5")**	3048 **(10'0")**



UNDERGROUND MINING — ROOM & PILLAR — ROOF SUPPORT CARRIERS

Model	Prefix	Exit Year	Bucyrus Model	Lift Capacity – No Ballast tonnes (Tons)	Machine Weight kg (lb)	Tram Speed Level Unloaded km/h (mph)	Engine Type	Engine Power kW (hp)
SH650 VFD	RS6	2014	VT650	45.0 **(50.0)**	46 040 — with battery **(101,500)**	6.6 **(4.1)**	Battery	N/A
SH650 D	RS5	2014	VT650D	45.0 **(50.0)**	45 359 kg **(100,000)**	13.7 **(8.5)**	Cummins C8.3	179 **(240)**
SH660 D	JE3	2014	FBL-55	55.0 **(60.625)**	48 000 — tanks full **(105,821)**	19.5 **(12.1)**	Cat C7 ACERT	171.5 **(230)**

LAND CLEARING

CONTENTS

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Land clearing must be treated more as an art than a science because production rates and methods vary greatly from one area to another. This section deals with the many variables in clearing and includes methods, equipment and procedures to determine productivity rates.

VARIABLES AFFECTING CLEARING OPERATIONS

Vegetative Growth — Factors affecting production and therefore cost, include the number of trees, size of trees, wood density, root systems, vines and undergrowth. These factors can be estimated by a "tree-count" as discussed under "Job Survey."

End Use of Land — Since different end uses require different degrees of clearing (i.e. highways, dams, tree crops, row crops, etc.), this is one of the most important factors to consider in choosing the proper clearing method and equipment.

Soil Conditions or Bearing Capacity — Factors affecting clearing operations include topsoil depth, soil type, moisture content, and the presence of rocks and stones.

Topography — Grade and terrain factors such as steep slopes, ditches, swampy areas, boulders and even ant hills greatly affect the normal operation of some equipment.

Rainfall and Climate — Usually all phases of land clearing from cutting to burning are concerned to some degree with temperature changes and the amount of rainfall during the clearing operation.

Job Specifications — Specifications dictate the degree of clearing to be done, area size, completion dates, method of debris disposal, soil conservation and other factors which affect method and equipment selection.

JOB SURVEYS

Knowledge of rainfall and climate, end use of the land, and job specifications can be obtained from records, surveys, engineering studies, and written specifications. You should personally review the land to be cleared to gain other necessary and valuable information.

The survey should include a study of general topography and soil conditions. Note such problem factors as hills, rocks, or swamps which would significantly affect production or which would require special treatment.

Cruise the area to be cleared and determine the acreage of each vegetative type (i.e. upland woods, low timberlands, swamps). Make at least three tree counts at random for each vegetation type. To conduct these counts, randomly locate two points 100 meters (328 feet) apart. Count and measure vegetative growth along a straight line between these points for a width of about 5 meters (16 feet) on both sides. This gives the population of 1/10 hectare (1/4 acre).

1. Density of vegetation less than 30 cm (12 in) diameter
   Dense — 1480 trees/hectare or more (600 trees/acre)
   Medium — 990-1480 trees/hectare (400-600 trees/acre)
   Light — less than 990 trees/hectare (400 trees/acre)
2. Presence of hardwoods expressed in percent
3. Presence of heavy vines
4. Average number of trees per hectare (2.47 acres) in each of the following ground level diameter size ranges:
   Less than 30 cm (1 ft)
   31 cm-60 cm (1-2 ft)
   61 cm-90 cm (2-3 ft)
   91 cm-120 cm (3-4 ft)
   121 cm-180 cm (4-6 ft)
5. Sum of diameter of all trees per hectare (2.47 acres) above 180 cm (6 ft) in diameter at ground level.

CLEARING METHODS AND EQUIPMENT

Methods for Initial Felling — There are several methods indicating the degree of clearing for initial felling and several types of equipment for use with each method. Equipment use in different size vegetation and different size areas is summarized in the table on the next page. This information should serve only as a rough guideline in selecting equipment. The economical land area for each type of equipment will vary with the capital cost of equipment and moving cost. It is also affected by whether there are alternate uses for equipment such as using tractors for other construction work or tillage.

Land Clearing Machines — Job size, severity of job such as tree size, and time limit to complete will influence machine selection. Some machines, such as purpose built forestry equipment and track-type tractors are more suited for this type of work than others, but imagination and resourcefulness can allow the use of other types of machines in specific applications. For example, loaders are used more today in raking and piling operations than ever before.

Operator Protection and Machine Guarding — Daily production has been estimated to increase 20% when cab guards are used. Cabs designed specifically for forestry and clearing applications are available on purpose built Cat® forestry equipment. Auxiliary equipment manufacturers such as Rome™ offer after market guard packages for non-purpose built units.

Locally designed and manufactured guarding for non-purpose built equipment is required. The radiator, engine, and underside of the tractor must be well protected. Perforated hoods, screens, crankcase guards and hydraulic cylinder guards are generally recommended.

Generally speaking lower cost clearing can be done with purpose built forestry equipment and larger tractors if the amount of clearing involved is sufficient to merit the initial investment of purpose built or if the amount of clearing involved is sufficient to merit the initial investment in the bigger machine. In applications where track-type tractors are used, power shift transmissions should be standard due to the frequent direction changes. The direct drive transmission tractor is recommended when the tractor is used principally in constant drawbar work such as chaining or pulling a disc harrow. In most applications, a winch should also be considered on one of every three tractors in a fleet.

EQUIPMENT SELECTION TABLE

	UPROOTING	CUTTING AT OR ABOVE GROUND LEVEL	KNOCKING TO THE GROUND	INCORPORATING INTO THE SOIL
LIGHT CLEARING — Vegetation up to 5 cm (2 in) diameter				
Small areas 4 hectares (10 acres)	Bulldozer blade; forestry machine road builder arrangement	Wheel-mounted circular saws	Bulldozer blade; forestry machine road builder arrangement	Moldboard plows, disc plows, disc harrows
Medium areas 40 hectares (100 acres)	Bulldozer blade; forestry machine road builder arrangement	Heavy duty sickle mowers [up to 3.7 cm (1½ in) diameter] tractor-mounted circular saws, suspended rotary mowers; site prep tractor with brush cutter or mulcher attachment	Bulldozer blade, rotary mowers; flail-type rotary cutters; rolling brush cutters; forestry machine road builder arrangement; site prep tractor with brush cutter or mulcher attachment	Moldboard plows; disc plows, disc harrows; site prep tractor with mulcher attachment
Large areas 400 hectares (1000 acres)	Bulldozer blade, root rake, grubber, root plow, anchor chain drawn between two crawler tractors; rails	Site prep tractor with brush cutter or mulcher attachment	Rolling brush cutter; flail-type cutter; anchor chain drawn between two crawler tractors; rails; site prep tractor with brush cutter or mulcher attachment	Undercutter with disc; moldboard plows; disc plows; disc harrows; site prep tractor with mulcher attachment
INTERMEDIATE CLEARING — Vegetation 5 to 20 cm (2 to 8 in) diameter				
Small areas 4 hectares (10 acres)	Bulldozer blade; forest machine road builder arrangement	Wheel-mounted circular saws	Bulldozer blade; forestry machine road builder arrangement	Heavy-duty disc plow; disc harrow
Medium areas 40 hectares (100 acres)	Bulldozer blade; forest machine road builder arrangement	Tractor-mounted circular saws, single scissor type tree shears; site prep tractor with brush cutter or mulcher attachment; wheel/track feller buncher with high speed saw head; forest machine with directional felling head	Bulldozer blade, rolling brush cutter [up to 12 cm (5 in) diameter], rotary mower [up to 10 cm (4 in) diameter]; forestry machine road builder arrangement; Site prep tractor with brush cutter or mulcher attachment	Heavy-duty disc plow; disc harrow; site prep tractor with mulcher attachment
Large areas 400 hectares (1000 acres)	Shearing blade, angling (tilted) bulldozer blade, rakes, anchor chain drawn between two crawler tractors, root plow	Shearing blade (angling or V-type); site prep tractor with brush cutter or mulcher attachment; wheel/track feller buncher with high speed saw head; forest machine with directional felling head	Bulldozer blade, flail-type rotary cutter, anchor chain; site prep tractor with brush cutter or mulcher attachment	Bulldozer blade with duty harrow; site prep tractor with mulcher attachment

NOTE: The most economical size area for each type of equipment will vary with the relative cost of capital equipment versus labor. It is also affected by whether there are alternate uses for equipment such as using tractors for tillage.

EQUIPMENT SELECTION TABLE

	UPROOTING	CUTTING AT OR ABOVE GROUND LEVEL	KNOCKING TO THE GROUND	INCORPORATING INTO THE SOIL
LARGE CLEARING — Vegetation 20 cm (8 in) diameter or larger				
Small areas 4 hectares (10 acres)	Bulldozer blade; forest machine road builder arrangement; Forest machine with directional felling head	—	Bulldozer blade; forest machine road builder arrangement; forest machine with directional felling head	—
Medium areas 40 hectares (100 acres)	Shearing blade, angling (tilted), knockdown beam, rakes, tree stumper; forest machine road builder arrangement; forest machine with directional felling head	Shearing blade (angling or V-type), tree shear [up to 70 cm (26 in) softwood; 35 cm (14 in) hardwood], shearing blade — power saw combination; wheel/ track feller buncher with high speed saw head; forest machine with directional felling head	Bulldozer blade; forest machine road builder arrangement; forest machine with directional felling head	—
Large areas 400 hectares (1000 acres)	Shearing blade, angling (tilted), tree pusher, rakes, tree stumper, anchor chain with ball drawn between two crawler tractors; forest machine with directional felling head	Shearing blade (angling or V-type), shearing blade — power saw combination; wheel/track feller buncher with high speed saw head; forest machine with directional felling head	Anchor chain with ball drawn between two crawler tractors. [Use dozer blade for trees over 18 cm (7 in).]; forest machine road builder arrangement; forest machine with directional felling head	—

NOTE: The most economical size area for each type of equipment will vary with the relative cost of capital equipment versus labor. It is also affected by whether there are alternate uses for equipment such as using tractors for tillage.

PRODUCTION ESTIMATING

GENERAL — CONSTANT SPEED OPERATIONS

Production is the hourly clearing rate usually expressed in hectares or acres.

For many land clearing operations, production is calculated by multiplying the tractor speed by the width of cut and converting to hectares or acres per hour.

Metric system:

The base formula is:

$$\frac{\text{Width of cut (meters)} \times \text{speed (km/h)}}{10} = \text{hectares/h}$$

When an efficiency of 82.5% is used, the formula becomes:

$$\frac{\text{Width of cut (m)} \times \text{speed (km/h)} \times 0.825}{10} = \text{hectares/h}$$

English measure:

$$\frac{\text{Width of cut (ft)} \times \text{speed (mph)}}{43{,}560\ (\text{ft}^2)} = \text{acres/hr}$$

The American Society of Agricultural Engineers formula for estimating hourly production of a constant speed operation is based on 82.5% efficiency. With this efficiency, the formula becomes:

$$\frac{\text{Width of cut (ft)} \times \text{speed (mph)} \times 0.825}{43{,}560\ (\text{ft}^2)} = \text{acres/hr}$$

Width of cut is the effective working width of the equipment and may not be the same as its rated width. Working width should be measured on the job but can be estimated when necessary.

The actual machine speed can be determined by measuring the amount of time to travel a given distance. When using the metric system, the time to travel 16.7 meters or a multiple thereof, can be converted into kilometers per hour.

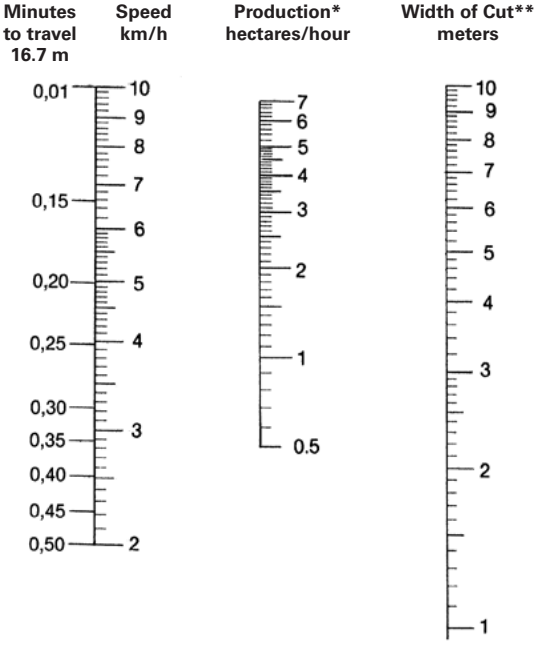
$$\frac{1.0}{\text{(Time in min. to travel 16.7 meters)}} = \text{speed (kmh)}$$

Since 88 ft/min. equals one mph, the lapsed time to travel 88 ft, or a multiple of 88 ft, can easily be converted into miles per hour.

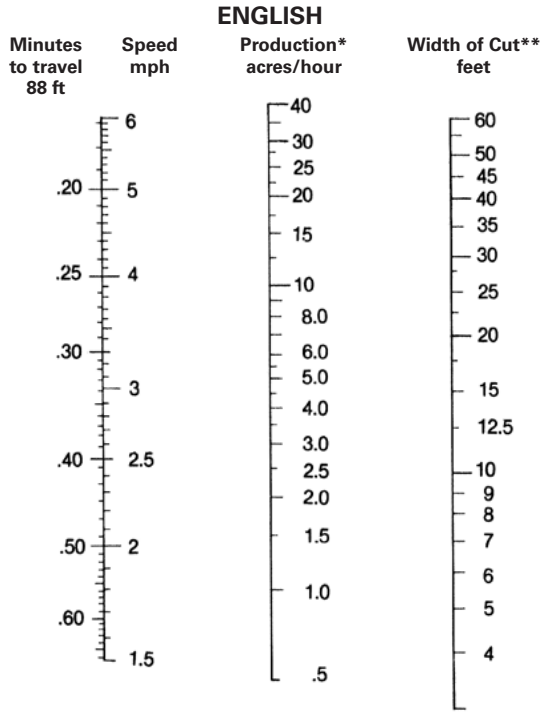
$$\frac{1.0}{\text{(Time in min. to travel 88 ft)}} = \text{speed (mph)}$$

The following nomographs in both the Metric and English systems convert speed and width of cut directly into acres or hectares per hour at 82.5% efficiency without the need for calculations.

METRIC



*Based on 82.5% efficiency.
**When width of cut exceeds 10 meters, use a multiple of the width of cut and increase production proportionately.



*Based on 82.5% efficiency.
**When width of cut exceeds 60 feet, use a multiple of the width of cut and increase production proportionately.

CUTTING PRODUCTION ESTIMATING

Most land clearing operations such as bulldozing, cutting, grubbing, raking and piling are not performed at constant speed. Because off-the-job production is difficult to estimate for these operations, Rome Industries has developed formulas for estimating cutting and piling time. These formulas take into consideration variable prime mover speeds through a factor, "B", the base time for each tractor to cover one hectare (2.47 acres) of light material.

To estimate **tractor cutting time per hectare** (2.47 acres) on a specific land clearing job, apply the factors shown in the following table, together with data obtained from the job survey, in the formula:

$$T = X\,[A(B)+M_1N_1+M_2N_2+M_3N_3+M_4N_4+DF]$$

where

T = Time per hectare (2.47 acres) in minutes
X = Hardwood or density factor affecting total time
A = Density or vine presence factor affecting base time
B = Base time for each tractor per hectare (2.47 acres)
M = Minutes per tree in each diameter range
N = Number of trees per hectare (2.47 acres) in each diameter range obtained from field survey
D = Sum of diameter in 30 cm (1 ft) increments of all trees per hectare (2.47 acres) above 180 cm (6 ft) in diameter at ground level obtained from field survey
F = Minutes per 30 cm (1 ft) of diameter for trees above 180 cm (6 ft) in diameter.

Hardwoods affect over-all or total time as follows:

75-100% hardwoods: Add 30% to total time (X = 1.3)
25-75% hardwoods: No change (X = 1.0)
0-25% hardwoods: Subtract 30% from total time (X = 0.7)

Production Factors for Felling with Rome K/G Blades

Tractor	Base Minutes per hectare (2.47 acres) "B"	Diameter Range 30-60 cm (1-2 ft) "M_1"	60-90 cm (2-3 ft) "M_2"	90-120 cm (3-4 ft) "M_3"	120-180 cm (4-6 ft) "M_4"	Dia. above 180 cm per 30 cm (6' per foot) "F"
165 hp	85	0.7	3.4	6.8	—	—
230 hp	58	0.5	1.7	3.3	10.2	3.3
305 hp	45	0.2	1.3	2.2	6.0	1.8
405 hp	39	0.1	0.4	1.3	3.0	1.0

Explanation of columns in table:

Tractor — Based on current model tractors (power shift when applicable) working on reasonably level terrain (below 10% grade) with good footing, no stones, average mixture of soft and hard woods. Tractor is in proper operating condition, blade is sharp, and properly adjusted.

Base Minutes — The base figures represent the number of minutes required for each tractor to cover a hectare (2.47 acres) of light material where no trees require splitting or other individual treatment. Time required is affected by the density of material less than 30 cm (1 ft) in diameter and the presence of vines.

a. dense — 1480 trees/hectare (600 or more trees/acre): Add 100% to base time (A = 2.0)
b. medium — 990-1480 trees/hectare (400-600 trees/acre): No change (A = 1.0)
c. light — less than 990 trees/hectare (400 trees/acre): Subtract 30% from total time (A = 0.7)

Presence of heavy vines: Add 100% to base time (A=2.0). *Very* heavy vines add 300% to base time. (A=3.0)

Dia. Range — M_1 represents minutes required to cut trees from 31-60 cm (1-2 ft) in diameter at ground level.

M_2 same for trees 61-90 cm (2-3 ft) diameter.

M_3 same for trees 91-120 cm (3-4 ft) diameter.

M_4 same for trees 121-180 cm (4-6 ft) diameter.

For Dia. above 180 cm (6 ft) — The figures in this column represent size the number of minutes required per 30 cm (1 ft) of diameter for each tractor to cut trees above 180 cm (6 ft) in diameter. Thus, to fell a 240 cm (8 ft) diameter tree would require 8 × 1.8 or approximately 14.4 minutes with a D8T.

Example problem:

Calculate the felling production of a D8T with K/G Blade in these conditions: reasonably level terrain, firm ground, well drained, 85% hardwoods with heavy vines and the following average tree count per hectare (2.47 acre):

Diameter Range	Less than 30 cm (1 ft) "B"	31-60 cm (1-2 ft) "N_1"	61-90 cm (2-3 ft) "N_2"	91-120 cm (3-4 ft) "N_3"	121-180 cm (4-6 ft) "N_4"	Sum Dia's Above 180 cm (6 ft) "D"
Number of Trees	1100	35	6	6	4	488 cm (16 ft)

Solution:

$T = X\,[A(B)+M_1N_1+M_2N_2+M_3N_3+M_4N_4+DF]$

$T = 1.3\,[2.0\,(45)+0.2\,(35)+1.3\,(6)+2.2\,(6)+6\,(4)+16\,(1.8)]$

$= 1.3\,(90+7+7.8+13.2+24+28.8)$

$= 1.3\,(170.8)$

$= 222$ minutes/hectare (90 min/acre)

● ● ●

Where the job requires grubbing trees and stumps greater than 30 cm (1 ft) in diameter at the same time the trees are sheared, use the same basic procedure as defined above including the variables for the presence of hardwoods. After time per hectare (acre) in minutes has been determined, increase the over-all or total time by 25%.

Where the job requires re-entering the area (after all trees have been sheared) to remove stumps with a tilted shearing blade or stumper, increase the total time by 50%.

PILING PRODUCTION ESTIMATING

A procedure has also been developed for estimating piling production for a tractor equipped with a K/G blade or rake.

To estimate tractor hours per hectare (acre) on a specific land clearing job, apply the factors shown in the following table with data obtained from the job survey, in the formula:

$T = B+M_1N_1+M_2N_2+M_3N_3+M_4N_4+DF$

where

T = Time per hectare (2.47 acre) in minutes.

B = Base time for each tractor per hectare (2.47 acre).

M = Minutes per tree in each diameter range.

N = Number of trees per hectare (2.47 acre) in each diameter range obtained from field cruise.

D = Sum of diameter in 30 cm (1 ft) increments of all trees per hectare (2.47 acre) above 180 cm (6 ft) in diameter at ground level obtained from field cruise.

F = Minutes per 30 cm (1 ft) of diameter for trees above 180 cm (6 ft) in diameter.

Production Factors for Piling in Windrows*

Tractor	Base Minutes per hectare (2.47 acres) "B"	Diameter Range 30-60 cm (1-2 ft) "M_1"	60-90 cm (2-3 ft) "M_2"	90-120 cm (3-4 ft) "M_3"	120-180 cm (4-6 ft) "M_4"	Dia. above 180 cm per 30 cm (6' per foot) "F"
165 hp	157	0.5	1.0	4.2	—	—
230 hp	125	0.4	0.7	2.5	5.0	—
305 hp	111	0.1	0.5	1.8	3.6	0.9
405 hp	97	0.08	0.1	1.2	2.1	0.3

*May be used with most types of raking tools and angled shearing blade. Windrows to be spaced approximately 61 meters **(200 feet)** apart.

Explanation of columns in table:

Tractor — Production with tractor working alone based on current model tractors (power shift when applicable) working on reasonably level (below 10% grade) terrain with good footing, no stones, average mixture of soft and hard woods. The tractor is in proper operating condition. Decrease total time by 25-50% depending on the number and size of trees when using three or more tractors in combination.

Base Minutes — The base figures represent the number of minutes required for each tractor to cover a hectare (2.47 acres) of light material.

Dia. Range — M_1 represents minutes required to pile trees from 31-60 cm (1-2 ft) diameter at ground level.

M_2 same for trees 61-90 cm (2-3 ft) diameter.

M_3 same for trees 91-120 cm (3-4 ft) diameter.

M_4 same for trees 121-180 cm (4-6 ft) diameter.

For Dia. above 180 cm (6 ft) — The figures in this column represent for each tractor size the number of minutes required per 30 cm (1 ft) of diameter to pile trees above 180 cm (6 ft) in diameter. Thus, to pile a 240 cm (8 ft) diameter tree would require 8×0.9 or approximately 7.2 minutes with a D8T tractor.

Where the job requires piling of grubbed trees and stumps greater than 30 cm (1 ft) in diameter, use the same basic procedure defined above and then increase over-all or total time by 25%.

In dense small diameter brush with few or no large trees, or when cutting is vine entangled, reduce the base time by 30%.

Example problem:

Calculate the windrow piling production of a D7R Series 2 with Rake in level terrain, no grubbing, and average mixture of hardwoods and softwoods where the average tree count per hectare (2.47 acres) is:

Diameter Range	Less than 30 cm (1 ft) "B"	31-60 cm (1-2 ft) "N_1"	61-90 cm (2-3 ft) "N_2"	91-120 cm (3-4 ft) "N_3"	121-180 cm (4-6 ft) "N_4"	Sum Dia's Above 180 cm (6 ft) "D"
Number of Trees	1100	35	6	6	2	0

Solution:

$T = B+M_1N_1+M_2N_2+M_3N_3+M_4N_4+DF$
$= 125+0.4\ (35)+0.6\ (6)+2.5\ (6)+5.0\ (2)+[DF=0]$
$= 42.6$
$= 177.6$ minutes/hectare (72 min/acre)

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To find the number of machines required for each operation, use the formula:

Hr/hectare (acre) × number of hectares (acres) = number of machines needed*

*Average machine production for all operation in hr/hectare (acre).

To cost estimate each method or phase of operation, use this calculation:

Owning and Operating cost/hr × hr/hectare (acre) × number of hectares (acres) = cost

Because of the many variables that increase or decrease production, these formulas should be considered only as guidelines in arriving at a rough production estimate. This estimate should be tempered by personal judgment based on past experience and personal knowledge of the area.

MINING AND EARTHMOVING

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INTRODUCTION

This section explains the earthmoving principles used to determine machine productivity. It shows how to calculate production on-the-job or estimate production off-the-job.

ELEMENTS OF PRODUCTION

Production is the hourly rate at which material is moved. Production can be expressed in various units:

Metric

Bank Cubic Meters — BCM — bank m^3
Loose Cubic Meters — LCM — loose m^3
Compacted Cubic Meters — CCM — compacted m^3
Tonnes

English

Bank Cubic Yards — BCY — bank yd^3
Loose Cubic Yards — LCY — loose yd^3
Compacted Cubic Yards — CCY — compacted yd^3
Tons

For most earthmoving and material handling applications, production is calculated by multiplying the quantity of material (load) moved per cycle by the number of cycles per hour.

Production = Load/cycle × cycles/hour

The load can be determined by

1) load weighing with scales
2) load estimating based on machine rating
3) surveyed volume divided by load count
4) machine payload measurement system

Generally, earthmoving and overburden removal for coal mines are calculated by volume (bank cubic meters or bank cubic yards). Metal mines and aggregate producers usually work in weight (tons or tonnes).

Volume Measure — Material volume is defined according to its state in the earthmoving process. The three measures of volume are:

BCM (BCY) — one cubic meter (yard) of material as it lies in the natural bank state.

LCM (LCY) — one cubic meter (yard) of material which has been disturbed and has swelled as a result of movement.

CCM (CCY) — one cubic meter (yard) of material which has been compacted and has become more dense as a result of compaction.

In order to estimate production, the relationships between bank measure, loose measure, and compacted measure must be known.

Swell — Swell is the percentage of original volume (cubic meters or cubic yards) that a material increases when it is removed from the natural state. When excavated, the material breaks up into different size particles that do not fit together, causing air pockets or voids to reduce the weight per volume. For example to hold the same weight of one cubic unit of bank material it takes 30% more volume (1.3 times) after excavation. (Swell is 30%.)

$$1 + \text{Swell} = \frac{\text{Loose cubic volume for a given weight}}{\text{Bank cubic volume for the same given weight}}$$

$$\text{Bank} = \frac{\text{Loose}}{(1 + \text{Swell})}$$

$$\text{Loose} = \text{Bank} \times (1 + \text{Swell})$$

Example Problem:

If a material swells 20%, how many loose cubic meters (loose cubic yards) will it take to move 1000 bank cubic meters (1308 bank cubic yards)?

Loose = Bank × (1 + Swell) =
1000 BCM × (1 + 0.2) = 1200 LCM
1308 BCY × (1 + 0.2) = 1570 LCY

How many bank cubic meters (yards) were moved if a total of 1000 loose cubic meters (1308 yards) have been moved? Swell is 25%.

Bank = Loose ÷ (1 + Swell) =
1000 LCM ÷ (1 + 0.25) = 800 BCM
1308 LCY ÷ (1 + 0.25) = 1046 BCY

Load Factor — Assume one bank cubic yard of material weighs 3000 lb. Because of material characteristics, this bank cubic yard swells 30% to 1.3 loose cubic yards when loaded, with no change in weight. If this 1.0 bank cubic yard or 1.3 loose cubic yards is compacted, its volume may be reduced to 0.8 compacted cubic yard, and the weight is still 3000 lb.

Instead of dividing by 1 + Swell to determine bank volume, the loose volume can be multiplied by the load factor.

If the percent of material swell is known, the load factor (L.F.) may be obtained by using the following relationship:

$$\text{L.F.} = \frac{100\%}{100\% + \%\ \text{swell}}$$

Load factors for various materials are listed in the Tables Section of this handbook.

To estimate the machine payload in bank cubic yards, the volume in loose cubic yards is multiplied by the load factor:

$$\text{Load (BCY)} = \text{Load (LCY)} \times \text{L.F.}$$

The ratio between compacted measure and bank measure is called shrinkage factor (S.F.):

$$\text{S.F.} = \frac{\text{Compacted cubic yards (CCY)}}{\text{Bank cubic yards (BCY)}}$$

Shrinkage factor is either estimated or obtained from job plans or specifications which show the conversion from compacted measure to bank measure. Shrinkage factor should not be confused with percentage compaction (used for specifying embankment density, such as Modified Proctor or California Bearing Ratio [CBR]).

Material Density — Density is the weight per unit volume of a material. Materials have various densities depending on particle size, moisture content and variations in the material. The denser the material the more weight there is per unit of equal volume. Density estimates are provided in the Tables Section of this handbook.

$$\text{Density} = \frac{\text{Weight}}{\text{Volume}} = \frac{\text{kg (lb)}}{\text{m}^3\,(\text{yd}^3)}$$

$$\text{Weight} = \text{Volume} \times \text{Density}$$

A given material's density changes between bank and loose. One cubic unit of loose material has less weight than one cubic unit of bank material due to air pockets and voids. To correct between bank and loose use the following equations.

$$1 + \text{Swell} = \frac{\text{kg/BCM}}{\text{kg/LCM}} \text{ or } \frac{\text{lb/BCY}}{\text{lb/LCY}}$$

$$\text{lb/LCY} = \frac{\text{lb/BCY}}{(1 + \text{Swell})}$$

$$\text{lb/BCY} = \text{lb/LCY} \times (1 + \text{Swell})$$

Fill Factor — The percentage of an available volume in a body, bucket, or bowl that is actually used is expressed as the fill factor. A fill factor of 87% for a hauler body means that 13% of the rated volume is not being used to carry material. Buckets often have fill factors over 100%.

Example Problem:

A 14 cubic yard (heaped 2:1) bucket has a 105% fill factor when operating in a shot sandstone (4125 lb/BCY and a 35% swell).

a) What is the loose density of the material?
b) What is the usable volume of the bucket?
c) What is the bucket payload per pass in BCY?
d) What is the bucket payload per pass in tons?

a) lb/LCY = lb/BCY ÷ (1 + Swell) = 4125 ÷ (1.35) = 3056 lb/LCY
b) LCY = rated LCY × fill factor = 14 × 1.05 = 14.7 LCY
c) lb/pass = volume × density lb/LCY = 14.7 × 3056 = 44,923 lb
   BCY/pass = weight ÷ density lb/BCY = 44,923 ÷ 4125 = 10.9 BCY
   or bucket LCY from part b ÷ (1 + Swell) = 14.7 ÷ 1.35 = 10.9 BCY
d) tons/pass = lb ÷ 2000 lb/ton = 44,923 ÷ 2000 = 22.5 tons

Example Problem:

Construct a 10,000 compacted cubic yard (CCY) bridge approach of dry clay with a shrinkage factor (S.F.) of 0.80. Haul unit is rated 14 loose cubic yards struck and 20 loose cubic yards heaped.

a) How many bank yards are needed?
b) How many loads are required?

a) $\text{BCY} = \frac{\text{CCY}}{\text{S.F.}} = \frac{10{,}000}{0.80} = 12{,}500 \text{ BCY}$

b) Load (BCY) = Capacity (LCY) × Load factor (L.F.) = 20 × 0.81
= 16.2 BCY/Load
(L.F. of 0.81 from Tables)

$$\text{Number of loads required} = \frac{12{,}500 \text{ BCY}}{16.2 \text{ BCY/Load}} = 772 \text{ Loads}$$

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Soil Density Tests — There are a number of acceptable methods that can be used to determine soil density. Some that are currently in use are:

Nuclear density moisture gauge
Sand cone method
Oil method
Balloon method
Cylinder method

All these except the nuclear method use the following procedure:

1. Remove a soil sample from bank state.
2. Determine the volume of the hole.
3. Weigh the soil sample.
4. Calculate the bank density kg/BCM (lb/BCY).

The nuclear density moisture gauge is one of the most modern instruments for measuring soil density and moisture. A common radiation channel emits either neutrons or gamma rays into the soil. In determining soil density, the number of gamma rays absorbed and back scattered by soil particles is *indirectly* proportional to the soil density. When measuring moisture content, the number of moderated neutrons reflected back to the detector after colliding with hydrogen particles in the soil is *directly* proportional to the soil's moisture content.

All these methods are satisfactory and will provide accurate densities when performed correctly. Several repetitions are necessary to obtain an average.

NOTE: Several newer methods have been successfully applied, along with weigh scales to determine volume and loose density of material moved in hauler bodies. These measurements include photogrammatic and laser scanning technologies.

FIGURING PRODUCTION ON-THE-JOB

Load Weighing — The most accurate method of determining the actual load carried is by weighing. This is normally done by weighing the haul unit one wheel or axle at a time with portable scales. Any scales of adequate capacity and accuracy can be used. While weighing, the machine must be level to reduce error caused by weight transfer. Enough loads must be weighed to provide a good average. Machine weight is the sum of the individual wheel or axle weights.

The weight of the load can be determined using the empty and loaded weight of the unit.

Weight of

load = gross machine weight – empty weight

To determine the bank cubic measure carried by a machine, the load weight is divided by the bankstate density of the material being hauled.

$$BCY = \frac{\text{Weight of load}}{\text{Bank density}}$$

Times Studies — To estimate production, the number of complete trips a unit makes per hour must be determined. First obtain the unit's cycle time with the help of a stop watch. Time several complete cycles to arrive at an average cycle time. By allowing the watch to run continuously, different segments such as load time, wait time, etc. can be recorded for each cycle. Knowing the individual time segments affords a good opportunity to evaluate the balance of the spread and job efficiency. The following is an example of a scraper load time study form. Numbers in the white columns are stop watch readings; numbers in the shaded columns are calculated:

Total Cycle Times (less delays)	Arrive Cut	Wait Time	Begin Load	Load Time	End Load	Begin Delay	Delay Time	End Delay
	0.00	0.30	0.30	0.60	0.90			
3.50	3.50	0.30	3.80	0.65	4.45			
4.00	7.50	0.35	7.85	0.70	8.55	9.95	1.00	10.95
4.00	12.50	0.42	12.92	0.68	13.60			

NOTE: All numbers are in minutes

This may be easily extended to include other segments of the cycle such as haul time, dump time, etc. Haul roads may be further segmented to more accurately define performance, including measured speed traps. Similar forms can be made for pushers, loaders, dozers, etc. *Wait Time* is the time a unit must wait for another unit so that the two can function together (haul unit waiting for pusher). *Delay Time* is any time, other than wait time, when a machine is not performing in the work cycle (scraper waiting to cross railroad track).

To determine trips-per-hour at 100% efficiency, divide 60 minutes by the average cycle time less all wait and delay time. Cycle time may or may not include wait and/or delay time. Therefore, it is possible to figure different kinds of production: measured production, production without wait or delay, maximum production, etc. For example:

Actual Production: includes all wait and delay time.

Normal Production (without delays): includes wait time that is considered normal, but no delay time.

Maximum Production: to figure maximum (or optimum) production, both wait time and delay time are eliminated. The cycle time may be further altered by using an optimum load time.

Example (English)

A job study of a Wheel Tractor-Scraper might yield the following information:

Average wait time	= 0.28	minute
Average load time	= 0.65	
Average delay time	= 0.25	
Average haul time	= 4.26	
Average dump time	= 0.50	
Average return time	= 2.09	
Average total cycle	= 8.03	minutes
Less wait & delay time	= 0.53	
Average cycle 100% eff.	= 7.50	minutes

Weight of haul unit empty — 48,650 lb
Weights of haul unit loaded —
Weighing unit #1 — 93,420 lb
Weighing unit #2 — 89,770 lb
Weighing unit #3 — 88,760 lb
271,950 lb;
average = 90,650 lb

1. Average load weight = 90,650 lb – 48,650 lb = 42,000 lb
2. Bank density = 3125 lb/BCY
3. $\text{Load} = \dfrac{\text{Weight of load}}{\text{Bank density}}$

$$= \frac{42{,}000 \text{ lb}}{3125 \text{ lb/BCY}} = 13.4 \text{ BCY}$$

4. Cycles/hr =

$$\frac{60 \text{ min/hr}}{\text{Cycle time}} = \frac{60 \text{ min/hr}}{7.50 \text{ min/cycle}} = 80 \text{ cycles/hr}$$

5. Production = Load/cycle × cycles/hr
(less delays) = 13.4 BCY/cycle × 8.0 cycles/hr
= 107.2 BCY/hr

Example (Metric)

A job study of a Wheel Tractor-Scraper might yield the following information:

Average wait time	= 0.28	minute
Average load time	= 0.65	
Average delay time	= 0.25	
Average haul time	= 4.26	
Average dump time	= 0.50	
Average return time	= 2.09	
Average total cycle	= 8.03	minutes
Less wait & delay time	= 0.53	
Average cycle 100% eff.	= 7.50	minutes

Weight of haul unit empty — 22 070 kg
Weights of haul unit loaded —
Weighing unit #1 — 42 375 kg
Weighing unit #2 — 40 720 kg
Weighing unit #3 — 40 260 kg
123 355 kg;
average = 41 120 kg

1. Average load weight = 41 120 kg – 22 070 kg = 19 050 kg
2. Bank density = 1854 kg/BCM
3. $\text{Load} = \dfrac{\text{Weight of load}}{\text{Bank density}}$

$$= \frac{19\,050 \text{ kg}}{1854 \text{ kg/BCM}} = 10.3 \text{ BCM}$$

4. Cycles/hr =

$$\frac{60 \text{ min/hr}}{\text{Cycle time}} = \frac{60 \text{ min/hr}}{7.50 \text{ min/cycle}} = 80 \text{ cycles/hr}$$

5. Production = Load/cycle × cycles/hr
(less delays) = 10.3 BCM/cycle × 8.0 cycles/hr
= 82 BCM/hrr